

可降阶的高阶微分方程

1) 形如 $y^{(n)} = f(x)$

$$\text{令 } y_1 = y^{n-1}$$

$$\frac{dy_1}{dx} = f(x) \quad \text{不断积分即可}$$

$$y''' = x + \sin x$$

$$y'' = \frac{1}{2}x^2 - \cos x + C_1$$

$$y' = \frac{1}{6}x^3 - \sin x + C_1x + C_2$$

$$y = \frac{1}{24}x^4 + \cos x + \frac{1}{2}C_1x^2 + C_2x + C_3$$

2) 形如 $f(x, y', y'') = 0$ 的不显含 y 的方程

$$\text{令 } y' = p(x) \quad y'' = \frac{dp}{dx} \quad f(x, p, \frac{dp}{dx}) = 0$$

$$\text{求 } xy'' + y' = 4x$$

$$\text{令 } y' = p(x) \quad y'' = \frac{dp}{dx}$$

$$x \frac{dp}{dx} + p = 4x \Rightarrow \frac{dp}{dx} + \frac{p}{x} = 4$$

$$\frac{dp}{dx} = -\frac{1}{x}p \Rightarrow d|\ln p| = -d|\ln x| \Rightarrow p = \frac{C}{x}$$

$$p = \frac{C(x)}{x} \Rightarrow \frac{dp}{dx} = -\frac{C(x)}{x^2} + \frac{C'(x)}{x} = -\frac{1}{x} \frac{C(x)}{x} + 4$$

$$\Rightarrow C'(x) = 4x \Rightarrow C(x) = 2x^2 + C \Rightarrow p = \frac{2x^2 + C}{x} = 2x + \frac{C_1}{x}$$

$$y = \int 2x + \frac{C_1}{x} = x^2 + C_1 \ln|x| + C_2$$

情形如 $f(y, y', y'') = 0$ 不显含自变量 x

$$\text{令 } y' = p(y)$$

$$y'' = \frac{dp}{dx} = \frac{dp}{dy} \frac{dy}{dx} = p \cdot \frac{dp}{dy}$$

$$f(y, p, p \frac{dp}{dy}) = 0$$

$$\frac{dy}{dx} = \frac{1}{p(y)} \Rightarrow x = \int \frac{1}{p(y)} dy$$

解 $y'' = \frac{(y')^2}{y + (y')^2}$ 满足初始条件 $y|_{x=0} = 1, y'|_{x=0} = 1$

$$\text{令 } y' = p(y), y'' = p \frac{dp}{dy}$$

$$\frac{dp}{dy} = \frac{p}{y + p^2} \Rightarrow \frac{dy}{dp} - \frac{1}{p} y = p$$

$$\text{通解 } y = p(C_1 + p) \text{ 代入 } y|_{x=0} = 1, y'|_{x=0} = 1$$

$$\Rightarrow C_1 = 0 \Rightarrow y = p^2$$

$$\frac{dy}{dx} = \sqrt{y} \Rightarrow x = 2\sqrt{y} + C_2 \text{ 代入 } y|_{x=0} = 1, y'|_{x=0} = 1$$

$$\Rightarrow C_2 = -2$$

$$x = 2\sqrt{y} - 2$$

二阶线性微分方程

$$y'' + p(x)y' + q(x)y = f(x)$$

设 $y_1(x), y_2(x), \dots, y_n(x)$ 为 $[a, b]$ 上的 $n-1$ 阶可微函数, 行列式

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) & \dots & y_n(x) \\ y_1'(x) & y_2'(x) & \dots & y_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & \dots & y_n^{(n-1)}(x) \end{vmatrix}$$

称为 $y_1(x), y_2(x), \dots, y_n(x)$ 的朗斯基行列式

对于不全为0的 n 个常数 $C_1, C_2, \dots, C_n$ 使得

$$C_1 y_1(x) + C_2 y_2(x) + \dots + C_n y_n(x) \equiv 0 \quad \forall x \in [a, b]$$

则称函数组 $y_1(x), y_2(x), \dots, y_n(x)$ 在区间 $[a, b]$ 上线性相关

设 $y_1(x), y_2(x)$ 是二阶齐次线性微分方程的两个解

$$y = C_1 y_1(x) + C_2 y_2(x)$$

$$y_1'' + p(x)y_1' + q(x)y_1 = 0$$

$$y_2'' + p(x)y_2' + q(x)y_2 = 0$$

$$\Rightarrow y'' + p(x)y' + q(x)y$$

$$= (C_1 y_1'' + C_2 y_2'') + p(x)(C_1 y_1' + C_2 y_2') + q(x)(C_1 y_1 + C_2 y_2)$$

设 $y_1(x), y_2(x)$ 是方程的两个线性无关的解则

$$y = C_1 y_1(x) + C_2 y_2(x)$$

是方程的通解

只需 C_1, C_2 为两独立常数

$y_1(x), y_2(x)$ 是线性无关的两个解

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} \neq 0$$

$$\frac{D(y, y')}{D(C_1, C_2)} = W(x) \neq 0$$

求微分方程 $xy'' - y' + (1-x)y = 0$ 的通解

$y_1 = e^x$ 为原方程的一个解

设 $y_2 = y_2(x)$

$$y'' - \frac{1}{x}y' + \frac{1-x}{x}y = 0$$

$$W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & y_2 \\ e^x & y_2' \end{vmatrix} = W(x) e^{-\int_{x_0}^x \frac{1}{t} dt} = W(x_0) \cdot \frac{x}{x_0}$$

$$\Rightarrow e^x (y_2' - y_2) = C_1 x$$

$$\Rightarrow y_2' - y_2 = C_1 x \cdot e^{-x} \quad y_2 = e^x [C_2 + C_1 \int x e^{-2x} dx] = e^x [C_2 - C_1 (\frac{1}{2} x e^{-2x} + \frac{1}{4} e^{-2x})]$$

$$\text{取 } C_1 = -4 \quad C_2 = 0 \quad y_2 = 2x e^{-x} + e^{-x}$$

$$y = C_1 e^x + C_2 (2x e^{-x} + e^{-x})$$

例 10.14 设线性无关的函数 $y_i(x) (i = 1, 2, 3)$ 都是微分方程

$$y'' + p(x)y' + q(x)y = f(x)$$

的解, 则其通解为().

(A) $C_1 y_1 + C_2 y_2 + y_3$

(B) $C_1 y_1 + C_2 y_2 - (C_1 + C_2) y_3$

(C) $C_1 y_1 + C_2 y_2 - (1 - C_1 - C_2) y_3$

(D) $C_1 y_1 + C_2 y_2 + (1 - C_1 - C_2) y_3$

解析 由于原方程的任意两个线性无关解的差是其对应的齐次方程的解, 于是 $y_1 - y_3, y_2 - y_3$ 是对应的齐次方程的解, 令

$$k_1(y_1 - y_3) + k_2(y_2 - y_3) = 0,$$

整理得

$$k_1 y_1 + k_2 y_2 - (k_1 + k_2) y_3 = 0,$$

由于 y_1, y_2, y_3 线性无关, 所以 $k_1 = k_2 = -(k_1 + k_2) = 0$, 即得 $k_1 = k_2 = 0$, 这表明 $y_1 - y_3$ 与 $y_2 - y_3$ 线性无关, 因而对应的齐次方程的

通解为

$$y = C_1(y_1 - y_3) + C_2(y_2 - y_3).$$

原方程的通解为

$$y = C_1(y_1 - y_3) + C_2(y_2 - y_3) + y_3.$$

故应选择(D). 其他解答皆错.

$y_1^*(x)$ 是非齐次方程 (1) $y'' + p(x)y' + q(x)y = f(x)$ 的一个解.

$y_2^*(x)$ 是齐次方程 (2) $y'' + p(x)y' + q(x)y = 0$ 的一个解.

则 $y_1^*(x) + y_2^*(x)$ 是 (1) 的一个解.

$y^*(x)$ 是非齐次线性微分方程 (1) 的一个特解.

$C_1 y_1(x) + C_2 y_2(x)$ 是齐次线性微分方程的通解.

则 $y(x) = C_1 y_1(x) + C_2 y_2(x) + y^*(x)$ 是 (1) 的通解.

常数变易法

$$y'' + p(x)y' + q(x)y = 0 \quad \text{通解为 } y = C_1 y_1(x) + C_2 y_2(x)$$

$$\text{设 } y'' + p(x)y' + q(x)y = f(x) \quad \text{的解为 } y = C_1(x) y_1(x) + C_2(x) y_2(x)$$

$$y' = C_1(x) \cdot y_1' + C_2(x) \cdot y_2' + \boxed{C_1'(x) \cdot y_1(x) + C_2'(x) \cdot y_2(x)} = 0$$

$$y'' = C_1(x) \cdot y_1'' + C_2(x) \cdot y_2'' + C_1'(x) \cdot y_1' + C_2'(x) \cdot y_2' + (\dots)$$

$$\begin{aligned} \Rightarrow y'' + p(x)y' + q(x)y &= C_1(x) (y_1'' + p(x)y_1' + q(x)y_1) \\ &\quad + C_2(x) (y_2'' + p(x)y_2' + q(x)y_2) \\ &\quad + C_1'(x)y_1' + C_2'(x)y_2' \end{aligned}$$

$$C_1'(x)y_1' + C_2'(x)y_2' = f(x)$$

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} C_1'(x) \\ C_2'(x) \end{pmatrix} = \begin{pmatrix} 0 \\ f(x) \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} C_1'(x) \\ C_2'(x) \end{pmatrix} = \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ f(x) \end{pmatrix} = \frac{1}{y_1 y_2' - y_2 y_1'} \begin{pmatrix} y_2' & -y_2 \\ -y_1' & y_1 \end{pmatrix} \begin{pmatrix} 0 \\ f(x) \end{pmatrix}$$

$$\begin{cases} C_1'(x) = \frac{1}{y_1 y_2' - y_2 y_1'} (-y_2 f(x)) \\ C_2'(x) = \frac{1}{y_1 y_2' - y_2 y_1'} (y_1 f(x)) \end{cases} \Rightarrow \begin{cases} C_1(x) = \\ C_2(x) = \end{cases}$$

若 $C_1(x)$ $C_2(x)$ 满足

$$\begin{cases} C_1'(x) y_1(x) + C_2'(x) y_2(x) = 0 \\ C_1'(x) y_1'(x) + C_2'(x) y_2'(x) = f(x) \end{cases}$$

非齐次方程有特解 $y^* = C_1(x) y_1(x) + C_2(x) y_2(x)$

求 $xy'' - y' + (1-x)y = 4x^2e^x$ 的通解

$$xy'' - y' + (1-x)y = 0 \text{ 通解为 } y = C_1e^x + C_2(2xe^{-x} + e^{-x})$$

$$y'' - \frac{1}{x}y' + \frac{1-x}{x}y = 4xe^x$$

$$\begin{cases} C_1'(x)e^x + C_2'(x)(2xe^{-x} + e^{-x}) = 0 \\ C_1'(x)e^x + C_2'(x)(e^{-x} - 2xe^{-x}) = 4xe^x \end{cases}$$

$$\begin{cases} C_1'(x) = 2x+1 \\ C_2'(x) = -e^{2x} \end{cases} \Rightarrow \begin{cases} C_1(x) = x^2+x + C_1 \\ C_2(x) = -\frac{1}{2}e^{2x} + C_2 \end{cases}$$

$$y = (x^2+x+C_1)e^x + (-\frac{1}{2}e^{2x}+C_2)(2xe^{-x}+e^{-x})$$

二阶线性常系数微分方程

$$y'' + py' + qy = 0 \quad (2)$$

$$\text{设 } y = e^{\lambda x} \quad y' = \lambda e^{\lambda x} \quad y'' = \lambda^2 e^{\lambda x}$$

$$\Rightarrow \lambda^2 + p\lambda + q = 0 \quad (1)$$

只要 λ 满足 (1), 则 $y = e^{\lambda x}$ 为 (2) 的解

I. $\lambda_1 \neq \lambda_2 \in \mathbb{R}$ $y = C_1e^{\lambda_1 x} + C_2e^{\lambda_2 x}$ 为 (2) 的通解

$$\begin{vmatrix} e^{\lambda_1 x} & e^{\lambda_2 x} \\ \lambda_1 e^{\lambda_1 x} & \lambda_2 e^{\lambda_2 x} \end{vmatrix} = (\lambda_2 - \lambda_1) \cdot e^{(\lambda_1 + \lambda_2)x} \neq 0 \Rightarrow e^{\lambda_1 x}, e^{\lambda_2 x} \text{ 线性无关}$$

II. $\lambda_1 = \lambda_2 = \lambda$ $e^{\lambda x}$ 为 (2) 的解.

用刘维尔公式求另一个与之线性无关的解

$$W(x) = \begin{vmatrix} e^{\lambda x} & y_2 \\ \lambda e^{\lambda x} & y_2' \end{vmatrix} = c \cdot e^{-\int p dx} = c \cdot e^{-px}$$

$$\Rightarrow e^{\lambda x} y_2' - \lambda e^{\lambda x} y_2 = c \cdot e^{-px} \Rightarrow y_2' - \lambda y_2 = c \cdot e^{-\lambda x}$$

$$y_2 = c_1 x \cdot e^{\lambda x} + c_2 e^{\lambda x}$$

III. $\lambda_{1,2} = \alpha \pm i\beta$

$$\begin{aligned} y &= \tilde{c}_1 e^{\lambda_1 x} + \tilde{c}_2 e^{\lambda_2 x} = \tilde{c}_1 e^{\alpha x} (\cos \beta x + i \sin \beta x) + \tilde{c}_2 e^{\alpha x} (\cos \beta x - i \sin \beta x) \\ &= c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x \end{aligned}$$

知 y_1 求 y_2

设 $y_2 = v(x) \cdot e^{\lambda x}$

$$y_2' = v'(x) \cdot e^{\lambda x} + \lambda v(x) \cdot e^{\lambda x}$$

$$y_2'' = v''(x) \cdot e^{\lambda x} + 2\lambda v'(x) \cdot e^{\lambda x} + \lambda^2 v(x) e^{\lambda x}$$

$$\text{代 (2)} \quad (\lambda^2 + p\lambda + q) v(x) e^{\lambda x} + (2\lambda + p) v' e^{\lambda x} + v''(x) e^{\lambda x} = 0$$

$$\Rightarrow v''(x) = 0. \Rightarrow v(x) = c_1 x + c_2$$

求 $y'' - 3y' - 4y = 0$ 的通解.

$$\lambda^2 - 3\lambda - 4 = 0$$

$$\lambda_1 = -1 \quad \lambda_2 = 4$$

$$y = C_1 \cdot e^{-x} + C_2 e^{4x}$$

求 $y'' + 2y' + y = 0$ 满足 $y(0) = 2$ $y'(0) = 1$ 的解.

$$\lambda^2 + 2\lambda + 1 = 0$$

$$\lambda = -1 \quad \text{重根}$$

$$\text{通解为 } y = C_1 \cdot e^{-x} + C_2 x e^{-x}$$

$$\begin{cases} 2 = y(0) = C_1 \\ 1 = y'(0) = -C_1 + C_2 \end{cases} \Rightarrow \begin{cases} C_1 = 2 \\ C_2 = 3 \end{cases} \Rightarrow y = 2e^{-x} + 3xe^{-x}$$

求 $y'' - 2y' + 5y = 0$ 的通解

$$\lambda^2 - 2\lambda + 5 = 0 \quad \lambda = 1 \pm 2i$$

$$\text{通解为 } y = C_1 e^x \cos 2x + C_2 e^x \sin 2x$$

求 $y^{(4)} + 2y'' + y = 0$ 的通解

$$\lambda^4 + 2\lambda^2 + 1 = 0$$

$$\lambda = \pm i \text{ 为特征方程的二重根}$$

$$y = (C_1 + C_2 x) \cos x + (C_3 + C_4 x) \sin x$$

求 $y''' - 3y'' + 3y' - y = 0$ 的通解.

$$\lambda^3 - 3\lambda^2 + 3\lambda - 1 = 0$$

$$(\lambda - 1)^3 = 0 \quad \lambda = 1 \text{ 的三重根}$$

$$y = c_1 e^x + c_2 x e^x + c_3 x^2 e^x$$

非齐次线性常系数微分方程

$$y'' + py' + qy = f(x)$$

待定系数法

$$f(x) = (a_0 x^m + a_1 x^{m-1} + \dots + a_m) e^{\mu x}, \text{ 其中 } \mu \text{ 以及 } a_i (i=0, 1, \dots, m) \text{ 为实常数.}$$

$$\text{则有 } y^* = x^k (A_0 x^m + A_1 x^{m-1} + \dots + A_m) e^{\mu x} \text{ 的特解}$$

k 为 μ 是特征方程 $F(\lambda) = \lambda^2 + p\lambda + q = 0$ 根的重数

当 μ 非特征根时, $k=0$.

$$\text{当 } \mu=0 \text{ 时 } f(x) = a_0 x^m + a_1 x^{m-1} + \dots + a_m$$

